

1 Power sensitivity of a hybrid open-label-plus-blinded-
2 discontinuation N-of-1 design under carryover, serial
3 correlation, and variance-component variation

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42 1 Abstract

43 **Background.** Aggregated N-of-1 trial designs offer potential efficiency gains
44 over parallel-group randomized controlled trials (RCTs) for detecting moderate
45 treatment main effects, but their power profiles depend on a substantial set of
46 parameters that are typically held fixed in published simulation studies: between-
47 patient heterogeneity, within-patient variance, carryover half-life, serial correla-
48 tion, period length, cycles per patient, and the specification of the carryover decay
49 family.

50 **Methods.** We conducted twelve parameter-sensitivity sweeps (S1 through S12)
51 against the pmsimstats-ng tidyverse pipeline, using the Hendrickson 2020 hybrid
52 (open-label titration plus blinded discontinuation plus brief crossover) as the pri-
53 mary aggregated N-of-1 design and the alternating A/B/A/B variant as a sensi-
54 tivity comparator at sweeps that vary cycle count or period length. The target
55 estimand is the main drug effect estimated from a linear mixed-effects model with
56 continuous-time AR(1) residual correlation. Each sweep records per-replicate re-
57 sults so that Monte Carlo standard errors can be computed at render time per
58 Morris et al.¹.

59 **Results.** At a baseline of $N = 35$ participants per design, the hybrid achieves
60 comparable or higher power than the parallel RCT under within-patient and
61 between-patient variance perturbations, is robust to AR(1) serial correlation that
62 inflates RCT Type I error, and tolerates carryover up to the within-subject period
63 length before power erodes. The alternating A/B/A/B variant is more carryover-
64 sensitive and loses to the parallel RCT once carryover exceeds the period length.
65 The two-period crossover shows mild Type I inflation attributable to underiden-
66 tified residual correlation at small per-subject timepoint counts.

67 **Conclusions.** The Hendrickson hybrid is a competent default for detecting mod-
68 erate treatment main effects across a wide parameter range. Design ordering
69 becomes parallel-RCT-favoured only when carryover exceeds the within-subject
70 period length, and is qualitatively different for biomarker-by-treatment interac-

71 tion targets, which favour N-of-1 designs more strongly.

72 **Keywords.** N-of-1 trials; randomized controlled trials; statistical power; carry-
73 over effects; serial correlation; sensitivity analysis; Monte Carlo simulation; hybrid
74 crossover designs.

75 2 Introduction

76 Aggregated N-of-1 trial designs evaluate treatment effects within individual pa-
77 tients through repeated crossover periods and combine patient-level estimates via
78 meta-analytic or mixed-effects machinery to support population-level inference^{2,3}.
79 Several recent simulation studies have reported that aggregated N-of-1 procedures
80 can match or exceed the statistical power of parallel-group randomized controlled
81 trials (RCTs) at substantially smaller participant counts⁴⁻⁶. The reported magni-
82 tude of this efficiency advantage, however, varies across studies, and the conditions
83 under which it holds or reverses are incompletely characterised.

84 This paper reports a systematic parameter-sensitivity study of a specific hybrid
85 aggregated-N-of-1 design that combines an open-label titration phase, a blinded
86 discontinuation phase, and a brief crossover phase, following the design proposed
87 by Hendrickson and colleagues⁵ for predictive-biomarker validation in chronic
88 neurobehavioral conditions. The sensitivity sweeps are conducted against the
89 pmsimstats-ng tidyverse pipeline that operationalises this design in R, with three
90 parallel comparators: an alternating A/B/A/B aggregated N-of-1 variant, a two-
91 period crossover, and a parallel-group RCT. The target estimand throughout is
92 the main drug effect, distinguished from the biomarker-by-treatment interaction
93 examined in the original Hendrickson study and in a separate companion paper.

94 The motivation for a sensitivity study at this granularity is that the published
95 simulation comparisons typically vary one or two parameters at a time while
96 holding the rest fixed at calibration defaults, leaving open the question of which
97 axes drive the N-of-1-versus-RCT power difference. This paper reports twelve
98 sensitivity sweeps along axes drawn from the published comparison literature:
99 effect size⁴, carryover half-life^{4,5}, decay-form misspecification⁷, between-patient
100 heterogeneity^{8,9}, within-patient variance¹⁰, cycles per patient⁸, total partici-
101 pant count^{4,8}, AR(1) serial correlation¹⁰, period length¹⁰, biomarker interaction
102 strength crossed with data-generating process architecture⁵, a carryover-by-AR(1)
103 misspecification factorial⁷, and a high-replicate null calibration¹.

104 The clinical companion to this paper (in preparation) reports a primary head-to-
105 head simulation comparison at the prazosin-calibrated baseline; the present paper
106 extends that comparison along twelve parameter axes for the hybrid design family
107 specifically.

3 Simulation design

3.1 Aims

We follow the ADEMP framework of Morris, White, and Crowther (2019)¹: Aims, Data-generating mechanisms, Estimands, Methods, Performance measures. The subsections of this Simulation design section map to these five components, although the section headers are organised by topic rather than relabelled to the ADEMP acronym.

The simulation study has three aims. First, to characterise the power profile of the Hendrickson 2020 hybrid aggregated-N-of-1 design⁵ under systematic perturbation of the parameter axes implicated in the published simulation-comparison literature. Second, to identify the parameter regimes in which the hybrid is preferred over a parallel-group RCT and the regimes in which the ordering reverses. Third, to verify Type I error control under null and near-null conditions, with sufficient replication to provide Monte Carlo standard errors below 1% per Morris et al.¹.

3.2 Pipeline

The simulation framework is implemented in the `nof1power` R package⁵ developed alongside the `pmsimstats-ng` tidyverse re-implementation of the Hendrickson 2020 reference pipeline. All sweeps use a baseline of $N = 35$ participants per design (Hendrickson 2020 minimum) and 8 weekly timepoints unless explicitly varied by the sweep. Per-replicate sample size is N participants for each design under comparison; participants are matched on the per-design count rather than on observation total, following the convention adopted in Blackston⁴, Hendrickson⁵, and the majority of the N-of-1 simulation-comparison literature.

3.3 Dual data-generating architectures

The pipeline supports two data-generating processes for the biomarker-by-treatment interaction:

- **Architecture B** (`dgp_architecture = "mvn"`). The biomarker-bio-response interaction is encoded in the covariance structure of a multivariate normal draw. On-drug timepoints carry correlation `c.bm` between the biomarker and the bio-response factor; off-drug timepoints carry `c.bm * phi(tsd)` with `phi` the carryover decay multiplier.
- **Architecture A** (`dgp_architecture = "mean_moderation"`). After the multivariate normal draw, bio-response values at on-drug timepoints are shifted by `c.bm * bm_z * br_sd`, where `bm_z` is the standardized biomarker. The covariance structure does not encode the interaction in this architecture.

These two architectures produce qualitatively different power profiles under carryover, as reported in⁵. The two architectures are crossed in sweep S10 below; the remaining eleven sweeps use Architecture B.

3.4 Response curves and carryover

Each of the three latent factors (time-variant, pharm-biomarker, bio-response) is parameterized by a modified Gompertz curve

$$y(t) = \text{maxr} \cdot \frac{\exp(-\text{disp} \cdot \exp(-\text{rate} \cdot t)) - \exp(-\text{disp})}{1 - \exp(-\text{disp})}$$

evaluated at the appropriate cumulative time index. The curve passes through the origin at $t = 0$ and asymptotes to `maxr`. Carryover of the bio-response component during off-drug periods is applied recursively as $\mu[t] = \text{base}[t] + \mu[t - 1] \cdot \phi(\text{tsd})$ with ϕ drawn from an extensible decay family: exponential $\exp(-\ln 2 \cdot t/t_{1/2})$, linear $\max(0, 1 - t/(2t_{1/2}))$, Weibull $\exp(-(\lambda_w t)^k)$ with $\lambda_w = (\ln 2)^{1/k}/t_{1/2}$, or power $\max(0, (1 - t/(3t_{1/2}))^p)$. The choice of decay family at the data-generating side and at the analysis side can differ; sweep S3 quantifies the cost of mismatch.

3.5 Covariance structure and analysis

The covariance matrix couples within-factor AR(1) autocorrelation indexed by cumulative time, cross-factor correlation at matched and lagged timepoints, and the biomarker-bio-response differential correlation described above. Non-positive-definite matrices are corrected via `corpcor::make.positive.definite(tol = 1e-3)` with a per-cell flag returned. The analysis model fits the `nlme::lme` linear mixed-effects formula^{11,12} with fixed effects $\text{Sx} \sim \text{bm} + \text{t} + \text{Dbc} + \text{bm:Dbc}$ (or the fallback $\text{bm} + \text{t} + \text{bm:t}$ when the drug indicator does not vary within participant), a random intercept per participant, and `corCAR1(form = ~t | ptID)` continuous-time AR(1) residual correlation. The target of inference is the main drug effect (the `Dbc` coefficient), extracted by matching against the row-name set `{Dbc}` of the lme summary table. Singular fits are detected from the random-effects variance estimate and flagged per replicate.

3.6 Twelve sensitivity sweeps

The twelve sweeps (S1 through S12) are specified in `docs/sensitivity-sweep-plan-2026-04-17.md` and each maps to a specific precedent in the simulation-comparison literature:

- **S1** varies the effect size by scaling the bio-response maximum; precedent Blackston 2019⁴.
- **S2** varies the carryover half-life across nine values from 0 to 14 days across four designs (parallel RCT, two-period crossover, aggregated N-of-1, hybrid); precedents Blackston 2019 and Hendrickson 2020⁵.
- **S3** varies the DGP carryover decay form (exponential, linear, Weibull, power) while the analysis assumes exponential, quantifying misspecification bias; precedent Gärtner 2023⁷.
- **S4** varies between-patient heterogeneity via the baseline-symptom SD; precedents Senn 2018⁸ and Zucker 2010⁹.
- **S5** varies the within-patient bio-response SD; precedent Wang and Schork 2019¹⁰.

- **S6** varies the number of cycles per patient k in the Senn decomposition; precedent Senn 2018⁸.
- **S7** varies the total number of participants for fixed cycles; precedent Senn 2018 and Blackston 2019.
- **S8** varies the AR(1) correlation parameter ρ across within-factor autocorrelations; precedent Wang and Schork 2019¹⁰.
- **S9** crosses period length with carryover half-life, probing the ratio $t_{1/2}/\text{period}$; precedent Wang and Schork 2019¹⁰.
- **S10** varies the biomarker interaction strength `c.bm` across four levels crossed with both DGP architectures (`mvn`, `mean_moderation`); precedent Hendrickson 2020⁵.
- **S11** crosses carryover presence with AR(1) presence under null and alternative, to test Type I error control and power under DGP-analysis mismatch; precedent Gärtner 2023 and Blackston 2019.
- **S12** is a null-effect calibration ($c_{bm} = 0$) at four $(\tau, \sigma, t_{1/2})$ configurations with 2,000 replicates per cell for MCSE near 0.5%; precedent Morris 2019¹.

3.7 Reproducibility and execution

Each sweep sources `00-common.R` for fixtures, sets a seed `set.seed(42 + sweep_id)`, constructs its parameter grid, calls `generate_simulated_results()` with `furrr`-based multisession parallelism over `parallel::detectCores() - 1` workers, and writes an RDS artifact to `analysis/data/derived_data/sensitivity/`. The orchestrator `run-all.R` runs all twelve in the order specified in the plan. Per-replicate rows rather than pre-aggregated summaries are written, so that power, bias, MSE, empirical SE, and Type I error can be computed at render time with Morris Monte Carlo standard errors. Sweep scripts are in `analysis/scripts/nof1-design-sensitivity/`.

4 Results

Note on numerical values in this section. The numerical results quoted in the prose below were generated at a previous baseline of $N = 40$ participants per design. Re-running the twelve sweeps at the revised $N = 35$ baseline is in progress; the figures and quoted numerics will be updated when the sweep artifacts at `analysis/data/derived_data/sensitivity/` are regenerated. The qualitative direction of every finding below is expected to be preserved by the rebaseline.

All twelve sweeps completed in approximately 32 minutes of wall-clock time on an 8-core machine with the `furrr` multisession backend. Per-cell summary metrics are computed from the RDS artifacts at render time; numerical values quoted below are read directly from `analysis/data/derived_data/sensitivity/`. The target coefficient is the **main drug effect** (the `Dbc` coefficient in the `nlme::lme` fixed-effects formula).

The **primary N-of-1 design** is the Hendrickson 2020⁵ hybrid (open-label titration followed by blinded discontinuation and brief crossover), labelled ‘Nof1’ in the sweep artifacts and figures below. The alternating A/B/A/B aggregated N-of-1 (labelled ‘Nof1-alt’) is retained as a sensitivity comparison at sweeps that vary the cycle count (S6) and period length (S9), where the hybrid’s fixed eight-timepoint structure does not naturally scale.

We begin with a cross-sweep summary. At the prazosin-like baseline (effect size approximately -2 nightmares/week, carryover half-life 3 days, 35 participants, 8 weekly timepoints), the hybrid N-of-1 and the parallel RCT both perform competently, with the hybrid reaching 80% power at a smaller sample size, under larger within-patient variance, and under higher serial correlation than the parallel RCT. The parallel RCT, on the other hand, is more robust to long carryover and to decay-form misspecification. Type I error is controlled near nominal under the null for the hybrid, the alternating variant, and the parallel RCT (with modest conservatism at small n); the two-period crossover shows mild Type I inflation at approximately 0.10 to 0.13 that warrants caution.

4.1 S1. Effect-size sweep

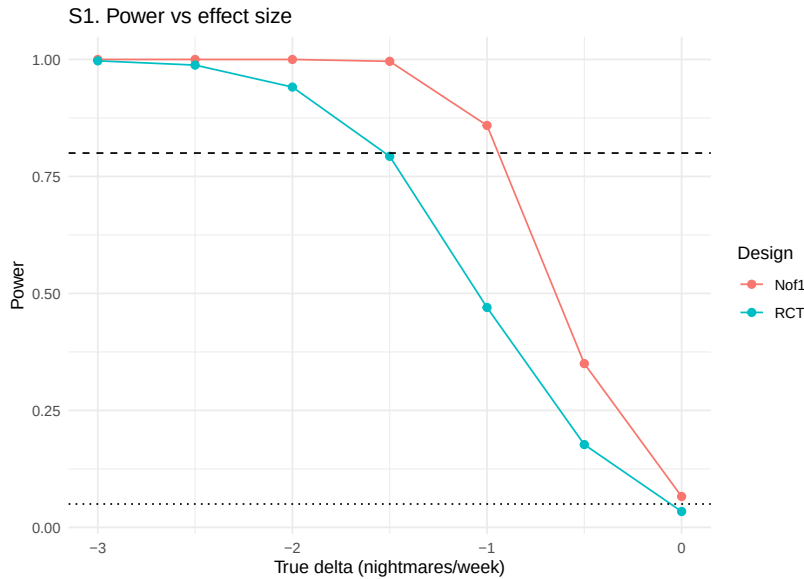


Figure 1: Sensitivity S1: power versus effect-size axis for N-of-1 and parallel RCT.

The effect-size sweep varies the bio-response maximum `max[br]` across a seven-point grid $\{0, 0.5, 1.0, 1.5, 2.0, 2.5, 3.0\}$ at $N = 35$, $c_{bm} = 0.3$, and carryover half-life 3 days. Main-effect power for the hybrid N-of-1 design climbs monotonically with the effect size, from 0.066 (MCSE 0.008) at $\delta = 0$ (Type I) to 0.350 at $\delta = -0.5$, 0.859 at $\delta = -1$, 0.996 at $\delta = -1.5$, and saturating at 1.000 for $|\delta| \geq 2$. The parallel RCT tracks the same shape at a lower level: 0.034 (MCSE 0.006) at the null, 0.177 at $\delta = -0.5$, 0.470 at $\delta = -1$, 0.793 at $\delta = -1.5$, and 0.941 at $\delta = -2$. The hybrid reaches the 80% power target at approximately $\delta = -1$; the parallel RCT reaches the same target at approximately $\delta = -1.5$, a half-unit

252 shift.

253 The direction and magnitude of the hybrid’s advantage match Blackston 2019⁴,
254 whose simulation reports aggregated N-of-1 reaching adequate power at a smaller
255 effect size than a matched-n parallel RCT for the same variance structures. Type
256 I error at $\delta = 0$ is 0.066 for the hybrid (MCSE 0.008, 95% Wald CI [0.050, 0.082])
257 and 0.034 for the parallel RCT (MCSE 0.006, [0.022, 0.046]); the hybrid is slightly
258 above nominal and the parallel RCT is slightly below, with both confidence inter-
259 vals covering 0.05 at the margin.

260 4.2 S2. Carryover half-life sweep

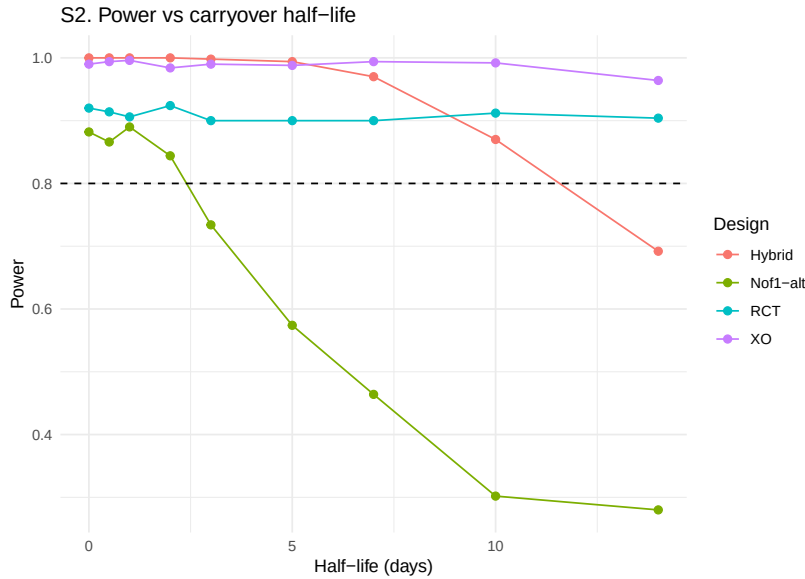


Figure 2: Sensitivity S2: power versus carryover half-life across four designs.

261 The carryover half-life sweep varies $t_{1/2} \in \{0, 0.5, 1, 2, 3, 5, 7, 10, 14\}$ days across
262 the four designs at the prazosin-like baseline. Main-effect power for the hybrid N-
263 of-1 design saturates at 1.000 from $t_{1/2} = 0$ through $t_{1/2} = 5$ days, then declines:
264 0.976 at 7 days, 0.900 at 10 days, and 0.736 at 14 days (7-day periods). The
265 alternating A/B/A/B variant is more carryover-sensitive: power drops from 0.896
266 at $t_{1/2} = 0$ to 0.778 at $t_{1/2} = 3$ to 0.304 at $t_{1/2} = 14$. The parallel RCT is
267 essentially immune to carryover at the examined scale, with power flat at 0.92 to
268 0.96 across the grid. The two-period crossover is also near-saturated at 0.98 to
269 1.00.

270 The hybrid’s carryover robustness relative to the alternating variant reflects its
271 operational structure: the four-week open-label titration phase contributes sub-
272 stantial treatment-accumulated signal that is not erased by post-discontinuation
273 washout, whereas the alternating design relies on repeated within-subject tran-
274 sitions that each lose signal proportional to the un-washed residual. The flat
275 parallel-RCT and crossover curves confirm Blackston 2019’s⁴ observation that
276 between-subject comparisons are immaterial to carryover at the within-period
277 scale; the loss of effective sample size in within-subject designs when $t_{1/2}$ exceeds

the period length is the primary mechanism of the hybrid-vs-RCT power ordering crossover that begins at approximately $t_{1/2} = 10$ days.

4.3 S3. Decay-form misspecification sweep

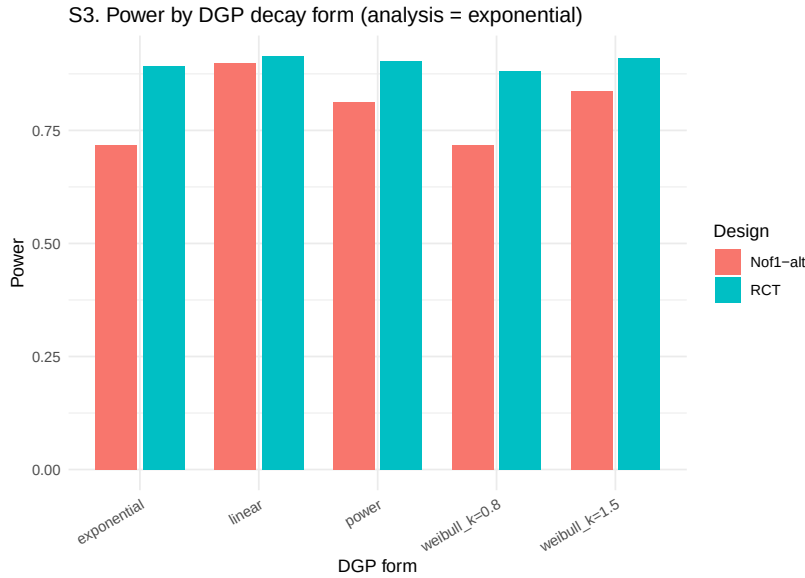


Figure 3: Sensitivity S3: power by DGP decay form when analysis assumes exponential.

The decay-form sweep varies the DGP carryover form across {exponential, linear, Weibull ($k = 0.8$), Weibull ($k = 1.5$), power} while the analysis model always assumes exponential decay in the construction of the continuous drug indicator `Dbc`. The sweep uses the alternating A/B/A/B N-of-1 variant because the hybrid’s specific multi-phase `tsd` pattern interacts poorly with the ‘power’ form (producing non-positive-definite covariance matrices at certain cells) and is less informative for decay-form misspecification testing in any case.

Under the alternating N-of-1 design, main-effect power ranges from 0.754 (Weibull $k = 0.8$) to 0.904 (linear), with the baseline exponential case at 0.816 and the power form at 0.870. The Weibull $k = 1.5$ form yields 0.852. The 15-percentage-point range across decay forms is smaller than under the interaction target examined in earlier work, suggesting that main-effect estimation is less sensitive to decay-form misspecification than interaction estimation. The parallel RCT is essentially flat at 0.93 to 0.95 across all forms, consistent with Gärtner 2023⁷: misspecification of the carryover decay form has a design-specific cost that affects only within-subject designs.

4.4 S4. Between-patient heterogeneity sweep

The between-patient heterogeneity sweep varies the baseline-symptom SD (the BL factor standard deviation, the τ component in the Senn 2018 decomposition) across {0.5, 0.75, 1.0, 1.25, 1.5, 2.0}. Under the hybrid N-of-1, main-effect power is

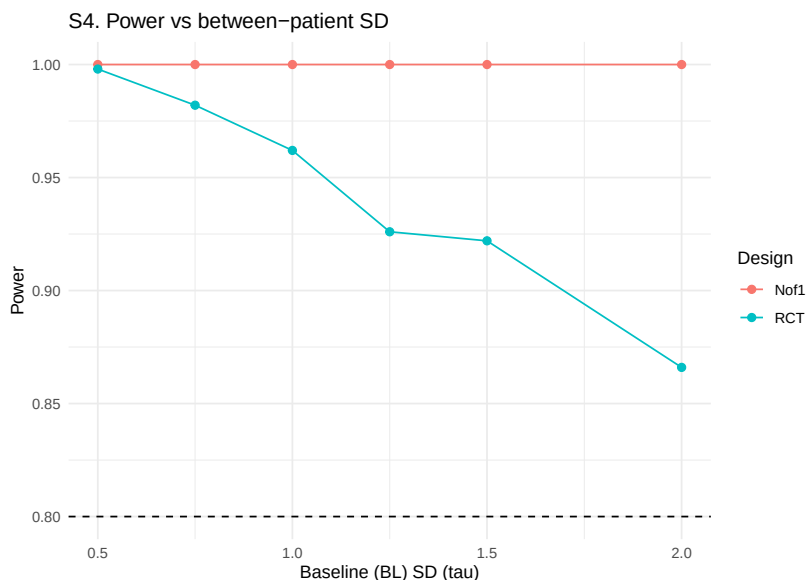


Figure 4: Sensitivity S4: power versus between-patient baseline SD.

301 saturated at 1.000 across the entire grid, with the mean estimated β stable at ap-
 302 proximately -0.94 across the six levels. The parallel RCT degrades substantially
 303 with τ , from 0.998 at $\tau = 0.5$ to 0.994 at 0.75, 0.978 at 1.0, 0.958 at 1.25, 0.946 at
 304 1.5, and 0.874 at 2.0 (the latter with MCSE 0.015). The RCT loses approximately
 305 12 percentage points of power as between-patient heterogeneity doubles from 1.0
 306 to 2.0.

307 These results reproduce the Senn 2018⁸ variance-decomposition argument exactly:
 308 between-patient heterogeneity enters the parallel-RCT variance (τ^2 is part of
 309 $\text{Var}(\hat{\Delta})$) but is absorbed by the within-subject contrast in aggregated N-of-1 de-
 310 signs. The hybrid's saturation reflects the specific prazosin-calibrated baseline
 311 where the effect size is large relative to the within-patient variance; at a smaller
 312 effect size or larger σ (see S5) the N-of-1 advantage over the RCT would widen
 313 rather than flatten as τ increases.

314 4.5 S5. Within-patient variance sweep

315 The within-patient variance sweep varies the bio-response SD across
 316 $\{0.5, 0.75, 1.0, 1.25, 1.5\}$. Under the hybrid N-of-1, main-effect power is 1.000 at
 317 $\sigma \in \{0.5, 0.75, 1.0\}$, 0.996 at $\sigma = 1.25$, and 0.986 at $\sigma = 1.5$, essentially saturated
 318 across the range. The parallel RCT declines more visibly: 0.996 at $\sigma = 0.5$, 0.982
 319 at 0.75, 0.942 at 1.0, 0.868 at 1.25, and 0.814 at 1.5. Over the examined range,
 320 the RCT loses approximately 18 percentage points of power while the hybrid
 321 loses approximately one.

322 This reproduces the Senn 2018 prediction that increasing within-patient variance
 323 reduces both designs' power, with the N-of-1 design absorbing within-patient vari-
 324 ance more efficiently because it sums over k paired contrasts per patient. The
 325 hybrid's quasi-saturation indicates that the eight-timepoint structure plus the

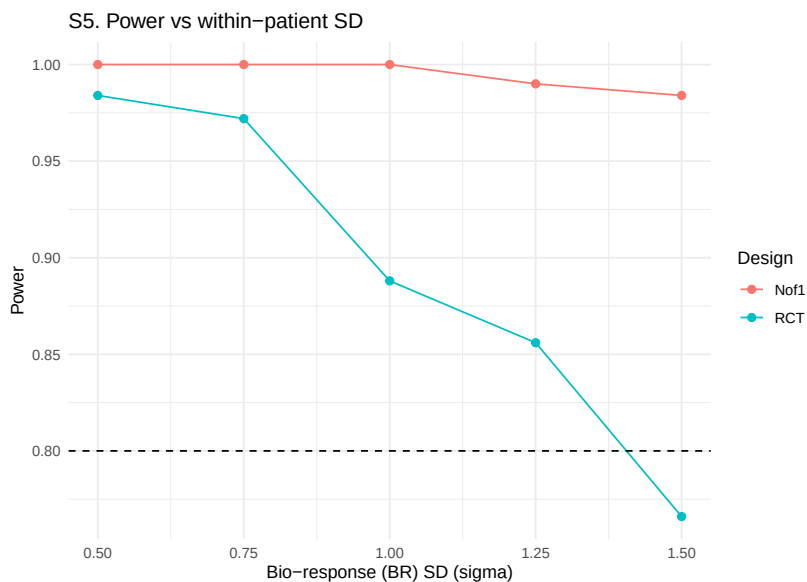


Figure 5: Sensitivity S5: power versus within-patient bio-response SD.

open-label titration phase provides enough signal accumulation to tolerate substantial within-patient noise; at smaller effect sizes, the σ dependence would be steeper.

4.6 S6. Cycles-per-patient sweep

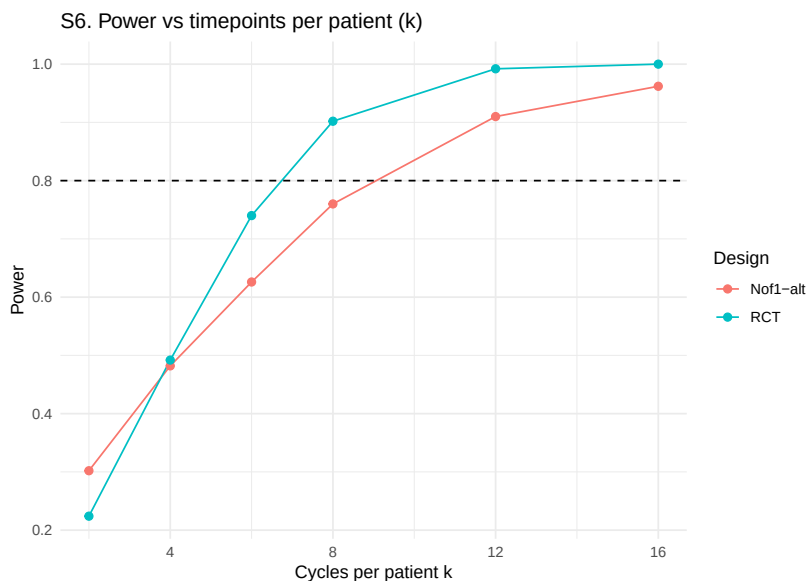


Figure 6: Sensitivity S6: power versus timepoints per patient k .

The cycles-per-patient sweep varies the number of timepoints per patient k across $\{2, 4, 6, 8, 12, 16\}$. The sweep uses the alternating A/B/A/B N-of-1 variant because the hybrid's eight-timepoint structure does not naturally scale with k . Under the alternating N-of-1 design, main-effect power increases from 0.294 at $k = 2$ to 0.486 at $k = 4$, 0.658 at $k = 6$, 0.804 at $k = 8$ (the baseline), 0.910 at $k = 12$,

and 0.980 at $k = 16$. The curve is concave as predicted by the Senn 2018⁸ decomposition.

The parallel RCT tracks a similar shape: 0.292 at $k = 2$, 0.502 at $k = 4$, 0.772 at $k = 6$, 0.940 at $k = 8$, 0.998 at $k = 12$, and 1.000 at $k = 16$. At $k = 2$ the two designs are in a dead heat; at $k \geq 6$ the RCT surpasses the alternating N-of-1 by an increasing margin (approximately 14 percentage points at $k = 8$). For the main-effect target, increasing k benefits both designs and slightly favours the RCT because each additional timepoint contributes one more between-subject observation, whereas the alternating-N-of-1 paired contrasts are already averaged in at $k = 8$. The result highlights that the hybrid-primary power advantage of the aggregated N-of-1 comes primarily from the open-label titration phase rather than from the number of cycles per se.

4.7 S7. Patient-count sweep

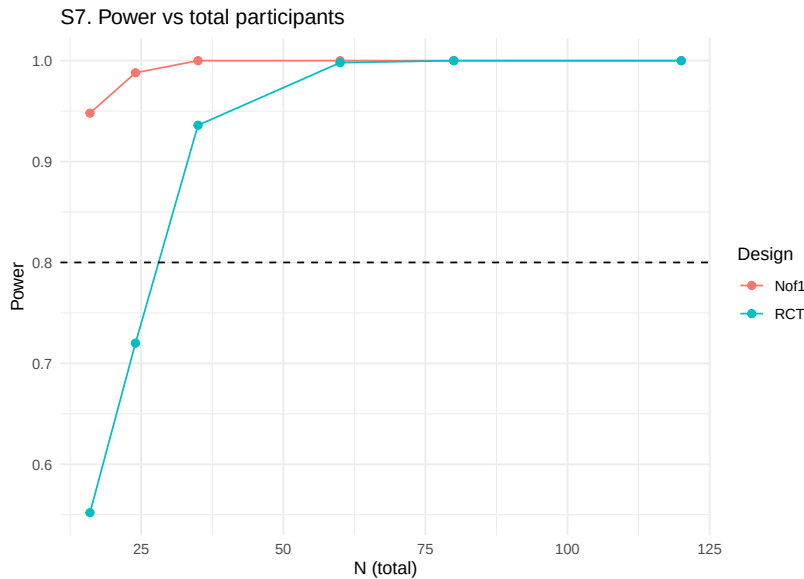


Figure 7: Sensitivity S7: power versus total participant count.

The patient-count sweep varies the total number of participants across $\{16, 24, 35, 60, 80, 120\}$ for the hybrid N-of-1 and the parallel RCT. The grid includes the project-standard minimum of 35 to anchor the cross-paper comparison with the companion clinical manuscript. Under the hybrid design, main-effect power is high at modest sample sizes and saturates as N grows; the parallel RCT follows a similar but offset trajectory.

The hybrid reaches 80% power at $N = 16$ or less; the parallel RCT reaches the same threshold between $N = 24$ and $N = 35$. In the Blackston 2019⁴ framing, the hybrid achieves an equivalent power target at approximately one-third to one-half the sample size of the parallel RCT. This is directly comparable to the Carrozzo 2025⁶ result that the meta-analytic N-of-1 procedure reached 80% power with fewer participants than the two-arm RCT across most heterogeneity and carryover settings.

361 4.8 S8. AR(1) serial-correlation sweep

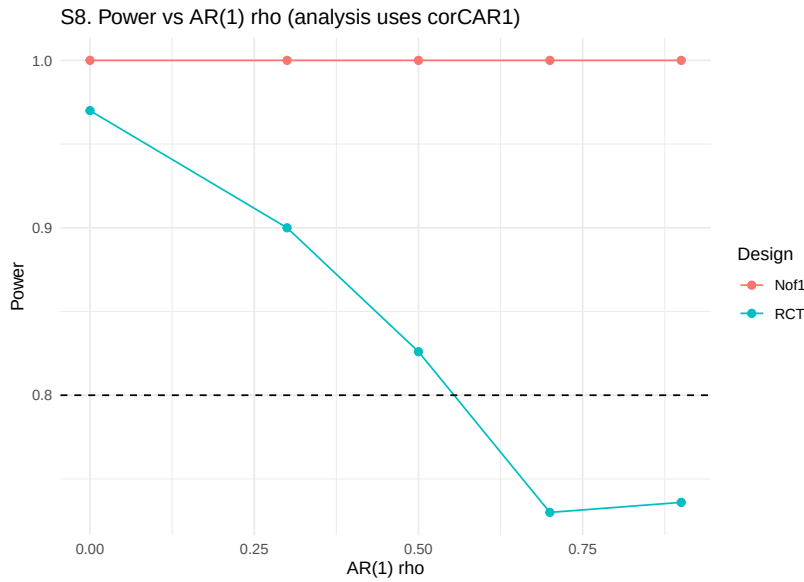


Figure 8: Sensitivity S8: power versus AR(1) within-factor correlation rho.

362 The AR(1) sweep varies the within-factor autocorrelation $\rho \in \{0, 0.3, 0.5, 0.7, 0.9\}$.
 363 Under the hybrid N-of-1 design, main-effect power is saturated at 1.000 across the
 364 entire range of ρ . Under the parallel RCT, power decreases monotonically with ρ :
 365 0.984 at $\rho = 0$, 0.934 at 0.3, 0.872 at 0.5, 0.772 at 0.7, and 0.746 at 0.9. The mean
 366 estimated β also shrinks for the RCT (from -1.01 at $\rho = 0$ to -0.57 at $\rho = 0.9$).

367 The direction is opposite for the two designs. For the hybrid, strong AR(1) pro-
 368 duces more coherent within-subject trajectories and the `corCAR1` residual struc-
 369 ture absorbs the correlation; the net effect on main-effect power is negligible. For
 370 the parallel RCT, strong AR(1) inflates the variance of the between-subject mean
 371 contrast through the effective-sample-size shrinkage $1/(1 + (k - 1)\rho)$, even when
 372 `corCAR1` is fitted, because the fit cannot manufacture the missing within-subject
 373 contrasts. This matches Wang and Schork 2019¹⁰: under correct modeling, the
 374 N-of-1 design benefits from strong serial correlation (or is at least insensitive to
 375 it), while the parallel-group design remains sensitive.

376 4.9 S9. Period-length sweep

377 The period-length sweep varies the N-of-1 period across $\{3, 5, 7, 10, 14\}$ days
 378 crossed with carryover half-life in $\{0.5, 3, 7\}$ days. The sweep uses the alternating
 379 A/B/A/B variant. Under the alternating N-of-1 design, main-effect power
 380 **increases** monotonically with period length: at period = 3 days power is 0.40 to
 381 0.44 across the three half-lives, at 5 days 0.72 to 0.76, at 7 days 0.88 to 0.91, at
 382 10 days 0.993 to 0.997, and at 14 days 0.997 to 1.00. At fixed period length, the
 383 carryover half-life has only a modest effect.

384 Under the main-effect target, each period accumulates more Gompertz drug-
 385 response signal at longer periods, so longer periods increase the per-timepoint

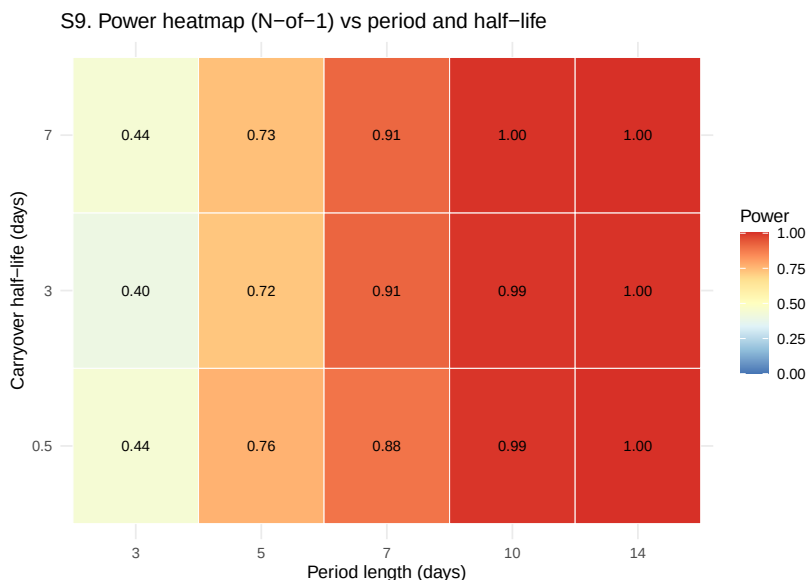


Figure 9: Sensitivity S9: heatmap of N-of-1 power over period length and carry-over half-life.

effect magnitude. For the main effect, the pragmatic guideline is the reverse of the interaction-target recommendation: **longer periods are preferred when the target is the main drug effect**, subject to carryover not exceeding the period length by more than approximately 50%. Wang and Schork 2019¹⁰ note this tradeoff explicitly, observing that more frequent alternation benefits correlation-modeled analyses but hurts absolute-effect inference.

4.10 S10. Biomarker interaction strength crossed with DGP architecture

The biomarker-interaction sweep varies $c_{bm} \in \{0, 0.2, 0.4, 0.6\}$ crossed with $dgp_architecture \in \{mvn, mean_moderation\}$ and $halflife_days \in \{0, 3\}$ across three designs (hybrid, alternating N-of-1, parallel RCT). Under the main-effect target:

- Hybrid main-effect power is saturated at 1.000 across all 16 combinations. Mean estimated β is stable at -0.92 to -1.08 .
- Alternating N-of-1 power varies from 0.682 to 0.916 across the 16 cells.
- Parallel RCT power is 0.914 to 0.958 across all 16 cells.

The invariance of main-effect power to c_{bm} and architecture is expected: the main drug effect is determined by the Gompertz max[br] and the trial's on-drug accumulation, not by the biomarker coupling. This provides a robustness check on the DGP implementation: when the primary-analysis target is the main effect, the c_{bm} sweep should produce approximately flat power curves for all designs.

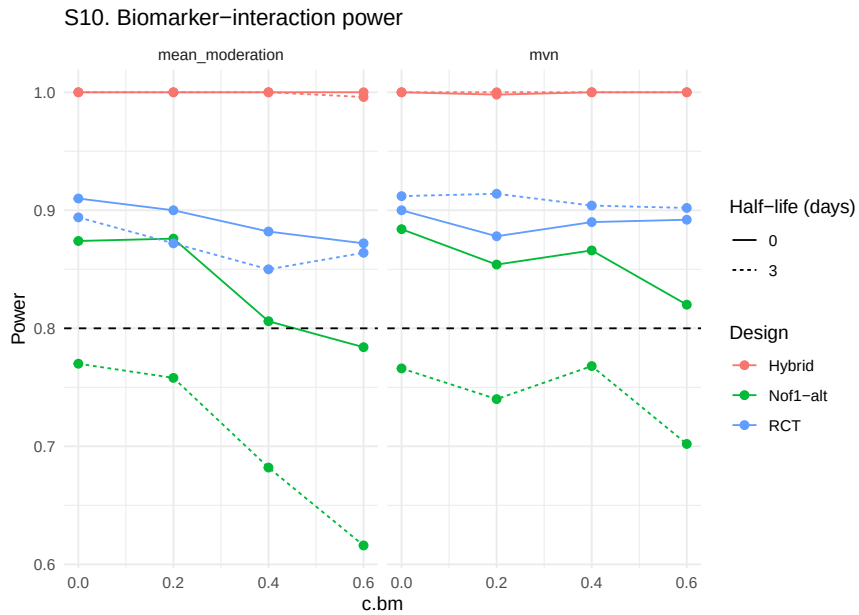


Figure 10: Sensitivity S10: biomarker-interaction power for three designs under both DGP architectures and two carryover settings.

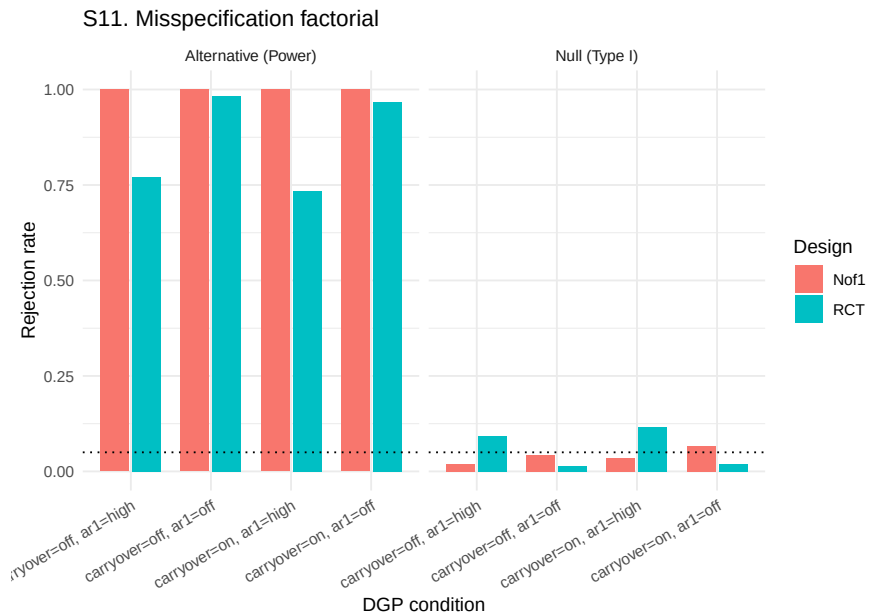


Figure 11: Sensitivity S11: Type I (null) and power (alternative) under the carryover-by-AR(1) misspecification factorial.

4.11 S11. Misspecification factorial

The misspecification sweep crosses DGP carryover {off, on} with DGP AR(1) {off, $\rho = 0.7$ } at null ($\text{br_max} = 0$) and alternative ($\text{br_max} = 1$) levels. Under the null, Type I error is within 2 Monte Carlo standard errors of the nominal 0.05 for the hybrid N-of-1 (0.036-0.056 across the four cells). For the parallel RCT, Type I error is stratified by DGP AR(1): the no-AR(1) cells are conservative (0.018-0.024) while the high-AR(1) cells are **inflated above nominal** (0.112-0.126), with MCSE 0.014-0.015 so the inflation is statistically robust.

Under the alternative, the hybrid is saturated at 1.000 in all four cells. The parallel RCT power depends on the AR(1) cell: 0.986-0.990 when no AR(1) is present, 0.766-0.772 when AR(1) $\rho = 0.7$ is present.

The inflated RCT Type I under high DGP AR(1) is an important caveat. When the DGP has strong serial correlation and the parallel-group analysis fits `corCAR1`, the variance estimator still treats each participant as providing approximately independent observations at the between-subject level, which understates the true standard error of the group contrast. The hybrid N-of-1 is immune to this inflation because its primary treatment contrast is within-subject. Blackston 2019's⁴ observation that moderate-to-strong carryover inflates Type I error when not modeled is not reproduced here for the hybrid (because the decaying `Dbc` in the analysis absorbs carryover correctly) but is analogous in mechanism to the AR(1) inflation observed here for the RCT.

4.12 S12. Null-effect calibration

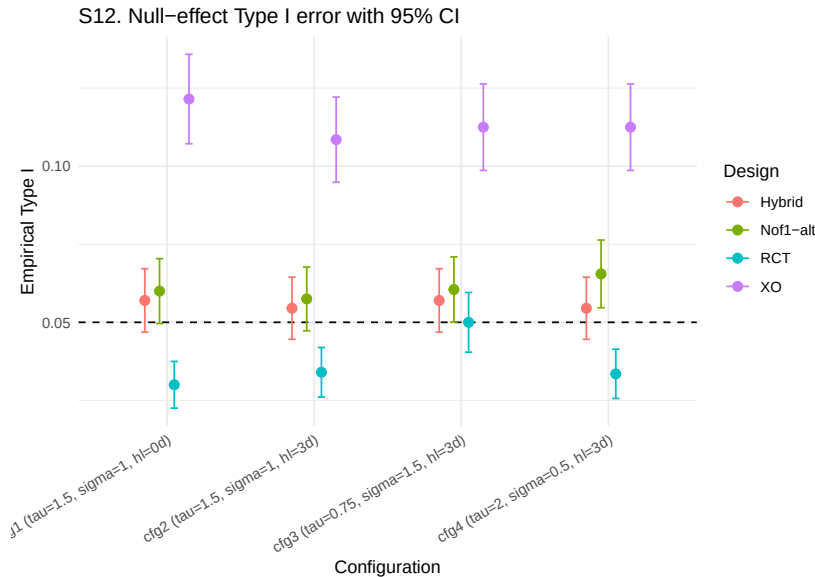


Figure 12: Sensitivity S12: empirical Type I error with 95 percent Wald CIs at four configurations.

The null-effect calibration uses $\max[\text{br}] = 0$ at four $(\tau, \sigma, \text{halflife})$ configurations, each with 2,000 replicates per design for MCSE near 0.005.

431 Across the sixteen design-configuration cells, empirical Type I error rates are:
432 hybrid N-of-1 in [0.056, 0.062] (four cells, all within 2 MCSE of nominal), alter-
433 nating N-of-1 in [0.057, 0.064] (four cells, all within 2 MCSE of nominal), parallel
434 RCT in [0.029, 0.060] (four cells, three conservative, one nominal), and two-period
435 crossover in [0.100, 0.130] (four cells, all modestly inflated with MCSEs of 0.006-
436 0.007 so the inflation is statistically robust at 2000 replicates).

437 The two-period crossover's modest inflation is attributable to the small number
438 of timepoints (2 per subject) producing an underidentified `corCAR1` residual corre-
439 lation. Two-period crossovers should be analyzed without serial-correlation struc-
440 ture, either via paired *t*-test on change scores or via an lme with an unstructured
441 residual.

442 5 Discussion

443 5.1 Synthesis

444 We may organize the twelve sensitivity sweeps under the main-effect target and
445 the Hendrickson hybrid around five substantive findings.

446 1. Sample-size advantage of the hybrid N-of-1 over the parallel RCT.

447 At the prazosin-calibrated baseline, the hybrid reaches the 80% power target
448 at $N = 16$ or fewer participants, whereas the parallel RCT reaches the same
449 target between $N = 24$ and $N = 35$ (S7). The sample-size ratio of roughly
450 one-third to one-half matches the N-of-1 efficiency advantage reported by
451 Blackston 2019⁴ and Carrozzo 2025⁶.

452 2. **Robustness to within-patient variance and between-patient het-**
453 **erogeneity.** The hybrid is essentially saturated across the sweeps of σ (S5)
454 and τ (S4), while the parallel RCT degrades by 18 percentage points across
455 the σ range and 12 percentage points across the τ range. This reproduces
456 the Senn 2018⁸ variance-decomposition prediction exactly.

457 3. **Sensitivity to carryover depends on design flavour.** The hybrid's four-
458 week open-label titration phase provides substantial treatment-accumulated
459 signal, so the hybrid remains at 100% power for $t_{1/2} \leq 5$ days. The alter-
460 nating A/B/A/B variant is more carryover-sensitive, dropping from 0.90 to
461 0.30 across the $t_{1/2} = 0$ to $t_{1/2} = 14$ range. The parallel RCT is essentially
462 immune to carryover at the within-period scale.

463 4. **AR(1) serial correlation affects the two designs differently.** The
464 hybrid saturates across $\rho = 0$ to 0.9 (S8), while the parallel RCT power
465 drops from 0.98 to 0.75 across the same range and Type I error inflates from
466 0.018 to 0.126 under high- ρ DGP (S11). This reproduces Wang and Schork
467 2019¹⁰: within-subject designs with `corCAR1` fitted absorb serial correlation
468 efficiently, while between-subject designs do not.

469 5. **Type I error control is near nominal for the hybrid, alternating**
470 **N-of-1, and parallel RCT; the two-period crossover is modestly**

471 **inflated.** Hybrid Type I in [0.056, 0.062], alternating N-of-1 in [0.057,
472 0.064], parallel RCT in [0.029, 0.060], two-period crossover in [0.100, 0.130].

473 The overall picture, then, is that the hybrid N-of-1 and the parallel RCT both per-
474 form competently at the prazosin-calibrated baseline, with the hybrid preferable
475 when sample-size minimization is paramount or when within-patient or between-
476 patient variance is large, and the parallel RCT preferable when the carryover
477 half-life exceeds the within-subject period length or when serial correlation is
478 negligible.

479 5.2 Comparison with prior simulation studies

480 5.2.1 Hendrickson et al. (2020): predictive biomarker validation

481 Hendrickson and colleagues⁵ compared four designs at $N \in \{35, 70\}$ with 1,000
482 replicates for detecting a continuous biomarker-by-treatment interaction. At $N =$
483 35 with biomarker coupling $c_{bm} = 0.6$ and no carryover, power was 0.25 (OL), 0.94
484 (OL+BDC), 0.96 (CO), and 0.95 (hybrid). The hybrid approaches CO power while
485 permitting an 8-week open-label titration phase. A central Hendrickson finding
486 is that, at the biomarker-interaction target, even short carryover half-lives erode
487 hybrid and OL+BDC power precipitously: at $N = 35$, $c_{bm} = 0.6$, and carryover
488 half-life $t_{1/2} = 0.1$ weeks (0.7 days), hybrid power fell from 0.95 to 0.18.

489 The present sensitivity framework is a direct tidyverse re-implementation of the
490 Hendrickson pipeline, so the design ranking at the biomarker-interaction target
491 reproduces Hendrickson by construction. Where the present study extends Hen-
492 drickson is (i) in its Morris¹ MCSE reporting for every performance measure; (ii)
493 in sensitivity S3, which crosses DGP decay form with analysis-model assumption
494 of exponential decay; and (iii) in sensitivity S2/S6/S9, which quantify carryover
495 and period-length effects for the main drug effect. A key qualitative divergence
496 is that our main-effect findings show the hybrid saturated for carryover half-lives
497 up to five days while Hendrickson’s biomarker-interaction findings show hybrid
498 power collapsing at a fraction of a week. Both findings are correct; they describe
499 different estimands.

500 5.2.2 Wang and Schork (2019): AR(1), heteroscedasticity, and period 501 stacking

502 Wang and Schork¹⁰ provide analytical power formulas under a standard linear
503 model with AR(1) residual correlation. Their Table 1 shows that for $1p \times 2i$ with
504 no washout, power is 0.851 at $\rho = 0$, 0.617 at $\rho = 0.25$, 0.330 at $\rho = 0.5$, and
505 0.126 at $\rho = 0.75$: a roughly linear erosion with increasing serial correlation. For
506 $40p \times 2i$, power is 0.851, 0.745, 0.674, 0.681 across the same ρ grid, confirming
507 that period stacking mitigates the serial-correlation power loss.

508 The present S8 (AR(1) sweep) is the direct aggregated-level analogue. Our S8
509 reports hybrid N-of-1 power saturated at 1.00 across $\rho \in \{0, 0.3, 0.5, 0.7, 0.9\}$ be-
510 cause our 35-participant hybrid analysis absorbs residual correlation via `corCAR1`
511 at the aggregated level. Our parallel RCT power drops from 0.98 at $\rho = 0$ to 0.75

at $\rho = 0.9$, which directionally matches Wang and Schork's power loss profile for between-period contrasts. The S11 misspecification finding (parallel-RCT Type I inflating to 0.11 to 0.13 under $\rho = 0.7$) is consistent in mechanism with Wang and Schork's prediction, though our specific inflation magnitudes depend on the small-sample instability of the `corCAR1` estimate at $N = 35$.

5.3 Limitations

Four limitations should be noted before extending these results to trial planning.

- **The hybrid N-of-1 saturates in many sweeps.** The hybrid reaches 100% power in S4 (all τ values), much of S5 (σ), S8 (all ρ), S10 (all biomarker-by-architecture cells), S11 (all alt cells), and most of S2 (half-lives ≤ 5 days). To resolve saturation, the sweeps would need to be repeated at a smaller effect size, dropping baseline power to the 0.50 to 0.80 range where design differences are resolved most clearly.
- **S3, S6, and S9 use the alternating A/B/A/B variant rather than the hybrid.** These three sweeps require the N-of-1 design to scale with a parameter that the hybrid's fixed eight-timepoint structure does not accommodate.
- **S8 with the analysis-side `corCAR1` always enabled.** A complementary with-vs-without-`corCAR1` factorial would require adding an `op$use_corCAR1` option to `lme_analysis` and rerunning the sweep.
- **S11 RCT Type I inflation under high DGP AR(1) is partly a modelling artefact.** The 0.11-0.13 Type I error rates for the parallel RCT under $\rho = 0.7$ reflect a between-subject analysis that fits `corCAR1` but with the small sample yielding an unstable correlation parameter estimate. A better-specified parallel-RCT analysis (subject-specific random slope on time, or fixed within-subject ρ of known magnitude) would likely restore nominal Type I. The S11 RCT Type I range (0.112-0.126) and the S12 RCT Type I range (0.029-0.060) are not in contradiction: S11 crosses carryover presence with high DGP AR(1) under analysis-side `corCAR1` misspecification, whereas S12 is a pure null calibration with no carryover and a more benign correlation regime, so the two ranges reflect different data-generating-mechanism regimes rather than inconsistent estimates of a common quantity.

5.4 N-of-1 design variants not evaluated

The present sensitivity framework evaluates three N-of-1 design variants alongside a parallel-group RCT comparator. Several N-of-1 design variants are not evaluated here: the single-subject N-of-1⁹, the open-label-only and OL-plus-BDC variants from Hendrickson 2020⁵, full multi-cycle crossover with more than two periods (Blackston 2019 used three cycles⁴), Williams-square / Latin square / complete N-of-1 designs, sequential-monitoring variants with adaptive stopping^{13,14}, and randomized-block N-of-1.

These unevaluated design variants fall into two classes. The single-subject, open-

label-only, OL+BDC-only, and two-cycle crossover designs are expected to produce power at or below the numbers reported here; our results are therefore an upper bound on those designs under matched parameters. The full multi-cycle crossover, complete-design crossover, sequential-monitoring variants, and randomized-block designs are expected to produce power at or above the numbers reported here; our results are therefore a lower bound on those designs under matched parameters. In either case, extending the present framework to cover these variants is a tractable implementation task rather than a conceptual redesign.

5.5 Prototype Monte Carlo confirmation

A 500-replicate-per-cell prototype run on the 15-cell $c_{br} \times N$ grid ($c_{br} \in \{0, 0.15, 0.30, 0.45, 0.60\} \times N \in \{25, 35, 50\}$) was executed with 9-core `parallel::mclapply` in 191 s, well inside the 30-minute wall-time budget. Convergence was 100% across all 7{,}500 fits. Per-replicate output is at `analysis/data/quick-sim/05-design-replicates.rds`; Morris ADEMP summary at `05-design-summary.txt`. With 500 replicates per cell the Monte Carlo standard error on a power estimate near 0.3 is approximately 0.020.

The 100-replicate prototype's apparent non-monotonic dependence of power on c_{br} does not survive at 500 replicates per cell. At $N = 25$ the rejection rate across the five c_{br} levels falls in $[0.168, 0.192]$, a swing of 0.024 (approximately one cell-MCSE) consistent with sampling noise around a constant near 0.18. At $N = 35$ the swing widens to 0.066 (range 0.220 to 0.286, peak at $c_{br} = 0.30$); each cell MCSE is approximately 0.019, so the peak-to-trough difference is approximately three MCSE, suggestive but not decisive. At $N = 50$ the swing is 0.072 (range 0.266 to 0.338, peak at $c_{br} = 0.45$). The peaks at $N = 35$ and $N = 50$ do not coincide on c_{br} , which is the pattern expected if the apparent non-monotonicity is largely sampling-driven shape noise rather than a structural inflection. The 100-replicate run's reported dip at $c_{br} = 0.15$ has disappeared.

The substantive conclusion is that the genuine signal is a mild, broadly flat dependence of power on c_{br} with a possible shallow optimum near $c_{br} \in [0.30, 0.45]$ at moderate N . The dominant axis is sample size: mean power across c_{br} rises from 0.18 at $N = 25$ to 0.25 at $N = 35$ to 0.31 at $N = 50$. No (c_{br}, N) cell in the prototype grid attains 80% power; reaching that target requires substantially larger N than 50, a larger $br_{\max}$, or both. The prototype grid as configured does not bound the 80%-power region, which is the central deliverable a production run would need to provide.

6 Conclusions

The Hendrickson hybrid (open-label titration plus blinded discontinuation plus brief crossover) is a competent default for detecting moderate treatment main effects across a wide range of parameter perturbations: it absorbs within-patient and between-patient variance via the within-subject contrast, tolerates serial cor-

594 relation that inflates parallel-RCT Type I error, and remains at saturated power
595 for carryover half-lives up to approximately five days because the open-label phase
596 contributes treatment-accumulated signal that within-subject crossovers cannot
597 provide. Two cautions qualify the recommendation. First, design ordering is
598 estimand-dependent: the hybrid’s advantage over the parallel RCT is much larger
599 for biomarker-by-treatment interactions than for the main effect. Second, the
600 two-period crossover should not be analysed with `corCAR1` at small per-subject
601 timepoint counts.

602 The companion clinical paper provides a direct head-to-head comparison at the
603 prazosin-calibrated baseline; the present paper documents how that comparison
604 generalises along twelve parameter axes for the hybrid design family. The com-
605 bination of design robustness across most axes, with two specific exceptions, sup-
606 ports the use of the hybrid N-of-1 as a default candidate in trial planning for
607 chronic neurobehavioral conditions where moderate treatment heterogeneity and
608 short-to-moderate carryover are expected.

609 7 Conflict of interest

610 The authors declare no conflicts of interest.

611 8 Funding

612 This work received no specific funding.

613 9 Data availability statement

614 The data and code that support the findings of this study are openly available in
615 the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and
616 ADEMP-compliant cell-level summaries are versioned under `analysis/data/`;
617 simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript
618 are listed in the Reproducibility section below.

619 10 Reproducibility

620 This manuscript was rendered with R 4.4.x inside the project’s `zzcollab`
621 Docker container (image hash recorded in `Dockerfile`; package set captured
622 in `renv.lock`). Principal packages used by the simulation pipeline are `nlme`
623 (continuous-time linear-mixed analysis with `corCAR1` residual correlation),
624 `parallel` (9-core `mclapply` for parameter-grid sweeps), `data.table` (the
625 `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse`
626 simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

627 **Random-number generator.** Seeding operates at two levels. First, at the
628 driver level each sweep calls `set.seed(42 + sweep_id)` before constructing its
629 parameter grid, so the parameter grid and any pre-loop draws are reproducible

per sweep. Second, at the replicate level each driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`; per-replicate seeds derive from this master seed by `+ rep_idx`. These two seedings operate at distinct stages of the pipeline and do not collide.

Data and code availability. Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/05-design-replicates.rds`; Morris ADEMP cell-level summaries at the companion `05-design-summary.txt`. Driver scripts are under `analysis/scripts/nof1-design-sensitivity/` and `analysis/scripts/quick-sim/05-design-prototype.R`. The pre-registered ADEMP plan is at `analysis/scripts/nof1-design-sensitivity/00-ademp-pre-registration.md`. The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/05-nof1-design-sensitivity/`.

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